

EU Semiconductors

BE Semiconductor Industries NV

Rating

Outperform

Price Target

 **BESI.NA**

320.00 EUR



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BESI: SDC Key Takeaways - HBM yield still being worked on but adoption trend unchanged

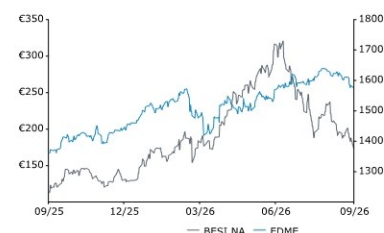
At Bernstein's 23rd SDC London, we met with BESI's CEO and IR team to discuss hybrid bonding adoption, advanced packaging trends, customer roadmaps, and the company's long-term outlook. Below are the key takeaways from the meeting:

- **Hybrid bonding adoption trend is going strong.**
 - **For Logic**, BESI's 50nm hybrid bonder ships its first system to TSMC this month. Throughput increases from ~1,500 to ~3,000 UPH, supporting a meaningful increase in ASPs.
 - **For HBM**, Samsung has been the most vocal memory player on HB adoption for HBM, while SK Hynix remains more cautious and is still testing HB, extending use of mass-reflow. BESI expects Hynix to adopt HB sooner rather than later, seeing timing as the key variable. Nvidia is also strongly advocating HB adoption for future HBM generations.
- **HB yield for memory still needs to be improved.** HB logic yields exceed 99.9% for current programs, including AMD, Broadcom and Apple M5-related production. HB yield for HBM is lower, typically in the 90% range and sometimes in the 70s for more challenging process flows.
- **HBM adoption trend is unchanged.** HBM4E HB volumes are expected to remain relatively modest, potentially limited to a single HB layer. BESI sees 2028/29 as the larger HB adoption ramp years. While memory adoption timing remains uncertain, CPO momentum is strong, with Nvidia already deploying HB and expressing significant confidence in the technology.
- **Service revenue** represents ~15% of sales and could increase toward ~20% over time, moving closer to the recurring revenue profile typically seen in front-end equipment companies.

Close Date	14 Sep 2026			
BESI.NA Close Price (EUR)	175.95			
Price Target (EUR)	320.00			
Upside/(Downside)	82%			
52-Week Range	328.40/112.55			
EDME	1,577.08			
FYE	Dec			
Div Yield	0.9%			
Market Cap (EUR) (M)	14,278			
EV (EUR) (M)	14,127			
Performance	YTD	1M	6M	12M
Absolute (%)	31.6	(24.6)	(4.9)	57.3
EDME (%)	7.2	(3.4)	6.6	14.2
Relative (%)	24.3	(21.3)	(11.5)	43.1

Source: Bloomberg, Bernstein estimates and analysis.

Price Performance, 1YR



Reported EPS	F25A	F26E	F27E	Financials	F25A	F26E	F27E	CAGR	Valuation Metrics	F25A	F26E	F27E
BESI.NA (EUR)	1.66	4.28	6.60	Revenues (M)	591.33	1,007	1,368	52.1%	Reported P/E (x)	106.0	41.1	26.6
				Operating Margin (%)	29.3	41.4	45.9	--				

Source: Bloomberg, Bernstein estimates and analysis.

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At Bernstein's 23rd SDC London, we met with BESl's IR team and CEO Richard Blickman to discuss HB adoption across logic and memory, HBM roadmaps, advanced packaging trends, customer qualification progress, the HB technology roadmap, and the company's long-term outlook. The discussions reinforced BESl's view that HB remains the inevitable long-term interconnect technology for both logic and memory due to its superior performance, power efficiency and thermal characteristics. Management also provided additional detail on HBM adoption timing, yield progress, next-generation 50nm platforms, service opportunities, and the competitive positioning that has allowed BESl to establish an early leadership position in HB.

HB Adoption in Memory (HBM)

- Management reiterated that the long-term adoption thesis for HB in HBM remains intact despite near-term delays. Samsung has been the most vocal memory manufacturer regarding HB adoption and is viewed as potentially the first-adopter of HB. SK Hynix remains more cautious and is still extending the use of mass-reflow technology, although BESl believes Hynix will adopt HB sooner rather than later. The company views timing as the main uncertainty rather than eventual adoption.
- Nvidia continues to push memory suppliers to offer HB-based HBM solutions due to their superior performance, lower power consumption and improved thermal characteristics. BESl expects HBM4E adoption volumes to remain relatively modest, potentially with HB used in only one layer, while 2028-29 should represent the more meaningful volume ramp. Even potential de-specification of future HBM roadmaps does not change BESl's long-term view, as more complex architectures such as zHBM and multi-layer planar stacking could actually increase the need for HB.
- While mass reflow and TCB continue to be extended because of their lower costs, management believes future increases in stack counts, shrinking geometries and thermal constraints will ultimately force the industry toward HB. Samsung and Micron have already purchased evaluation systems, while Hynix has recently begun testing the technology.

HB Yields: Logic vs Memory

- BESl highlighted the significant maturity gap between logic and memory applications. HB yields in logic are already highly mature, with yields exceeding 99.9% in production programs, including AMD, Broadcom, Apple M5-related products and Intel applications. Management is confident these levels can be replicated across future deployments at TSMC and Intel.
- HBM remains considerably more challenging, with yields often in the 90% range and occasionally in the 70% range for more difficult process flows. Management stressed that the challenge is not the bonder itself but rather the entire process flow, including wafer preparation, die preparation, materials and integration.
- According to BESl, moving from approximately 90% pre-production yields to 99.9% high-volume manufacturing yields typically requires one to two years of optimisation. Current delays in HBM adoption therefore appear to be driven primarily by process maturity rather than a lack of customer interest.

Service Revenue Opportunity

- Service currently accounts for approximately 15% of revenue and could increase towards 20% over time, moving BESl's business model closer to the recurring revenue profile typically seen in front-end semiconductor equipment companies.
- Because HB increasingly moves interconnect processes into the front-end environment, customer support requirements are substantially higher than in traditional assembly equipment. BESl expects growing demand for engineering support, process optimisation, training, spare parts and production assistance.

HB Product Roadmap and 50nm Platform

- BESl confirmed that the first 50nm HB system is being shipped to TSMC this month. The company has already tested four systems internally, including two focused on achieving 50nm accuracy and two using the same platform architecture to improve throughput on current-generation systems.
- Management indicated that the technology has largely moved from R&D into a pre-production phase. Qualification could continue for approximately another year, with broader production deployment expected from 2028 onwards.

- The new platform delivers a significant throughput improvement, increasing output from roughly 1,500 UPH to around 3,000 UPH. This productivity improvement supports a meaningful step-up in ASPs. BESI also confirmed that 20nm remains on the roadmap for 2030/31, while 10nm is viewed as a longer-term opportunity likely extending into the mid-to-late 2030s.

CPO, Photonics and Advanced Interconnects

- BESI remains highly constructive on CPO, photonics and advanced optical interconnects, all of which are becoming increasingly important as AI infrastructure scales. Management highlighted particularly strong momentum in pluggables, optical networking and photonics packaging applications.
- Nvidia was described as highly bullish on optical interconnect technologies and is already deploying HB in certain applications. BESI sees these markets as an additional structural growth driver for advanced packaging demand.
- More broadly, management noted that interconnect technology is becoming increasingly critical to overall system performance, helping explain why a growing number of industry participants are focusing on advanced packaging and back-end technologies.

HB Competitive Position and Market Share

- BESI believes its largest competitive advantage comes from being the clear technology leader in HB. The company has shipped over 180 systems, accumulated seven years of process experience and currently maintains sole-supplier positions at both Intel and TSMC.
- While customers generally prefer dual sourcing, BESI noted that TSMC has spent roughly a decade co-developing HB technology with the company and continues to rely exclusively on BESI despite evaluating competing alternatives, including Shibaura. To date, TSMC has not purchased any HB systems from competing suppliers, leaving BESI as the sole supplier. BESI also remains the sole supplier for Intel's relevant HB applications.
- Management expects market share to decline somewhat over time as Korean customers attempt to qualify alternative suppliers. However, the focus remains on maintaining technology leadership, being the process-of-record supplier and maximising returns rather than defending 100% market share.

Applied Materials Partnership

- Management repeatedly highlighted the importance of its partnership with Applied Materials. While TSMC was already a customer before the collaboration began, BESI believes the relationship with Applied has materially strengthened its ability to provide complete process solutions rather than standalone tools.
- As HB increasingly requires close integration between front-end and back-end manufacturing processes, Applied's expertise has become particularly valuable. BESI believes the partnership has contributed to process development, customer credibility and qualification efforts in both Korea and at Intel.
- The opening of Applied's EPIC Centre in October was highlighted as an important milestone. BESI HB equipment will be installed there, allowing customers to evaluate integrated process flows. Management views the partnership as critical to the next stage of HB adoption.

Traditional Packaging and Conventional Bonding

- While HB receives most investor attention, management highlighted that more than 80% of revenue still originates from non-HB products. BESI continues to benefit from strong demand in traditional packaging and maintains industry-leading margins in these businesses.
- Much of the growth is being driven by the CoWoS ecosystem, which requires significant deployment of mass-reflow equipment. Additional growth drivers include photonics, pluggables, connectors, 2.5D packaging, 3D packaging, mobile devices and recovering automotive demand.

- BESI reiterated its long-standing strategy of focusing on high-end, high-accuracy applications where differentiation through accuracy, yield and productivity can support premium pricing and attractive profitability.

Fluxless TCB

- BESI views fluxless TCB as an attractive complementary technology, particularly within memory applications where it may remain relevant even as HB adoption expands. The company currently has six customers evaluating or deploying fluxless TCB systems.
- Management noted that conventional TCB historically lacked the margin profile and differentiation required to meet BESI's investment criteria. Fluxless TCB offers a more attractive positioning and better economics.
- Discussions with Micron remain ongoing, and management suggested further developments could become clearer over the coming quarters.

HB Capacity and Order Visibility

- BESI has established dedicated HB manufacturing capacity with support from TSMC. Current cleanroom capacity supports approximately 20-25 HB systems per month, equivalent to 300 systems annually.
- Management stated that customers are heavily involved in capacity planning and that production can be expanded if demand accelerates. The company believes it is well positioned to support a broad industry transition towards HB.
- Order visibility remains strong, with customer roadmaps and qualification activity providing sufficient confidence for management to maintain its long-term HB guidance.

Margins and Business Model

- BESI reiterated its long-standing "margin over volume" strategy, focusing exclusively on high-end applications rather than pursuing market share in lower-margin categories. Management believes this approach has consistently differentiated the company from peers.
- HB is expected to be margin accretive, with GM around 70%, while the increasing mix of service revenue should further improve the business profile over time. Management therefore expects gross margins to continue trending higher as HB adoption increases.
- BESI also sees substantial operating leverage potential over the coming years and does not expect R&D spending to increase materially. As a result, R&D intensity should decline as a percentage of revenue as the company scales.

INVESTMENT IMPLICATIONS

We rate BESI Outperform with a PT of 320€

BERNSTEIN TICKER TABLE

			14 Sep 2026		TTM	Reported EPS				Reported P/E (x)		
Ticker	Rating	Cur	Closing Price	Price Target	Rel. Perf.							
BESI.NA (Besi)	O	EUR	175.95	320.00	43.1%	EUR	1.66	4.28	6.60	106.0	41.1	26.6
EDME			1,577.08									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

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VALUATION METHODOLOGY**BE Semiconductor Industries NV**

For Besi, we set a €320 price target using a P/E multiple of 40x against our forward Q5-Q8 EPS estimate.

RISKS**BE Semiconductor Industries NV**

Downside risks to our price target for Besi include short-term — lower than expected or delayed adoption of hybrid bonding and a prolonged delay in the recovery of the mainstream assembly market — losing market share in hybrid bonding or lack of take-up of Besi's TCB offering.

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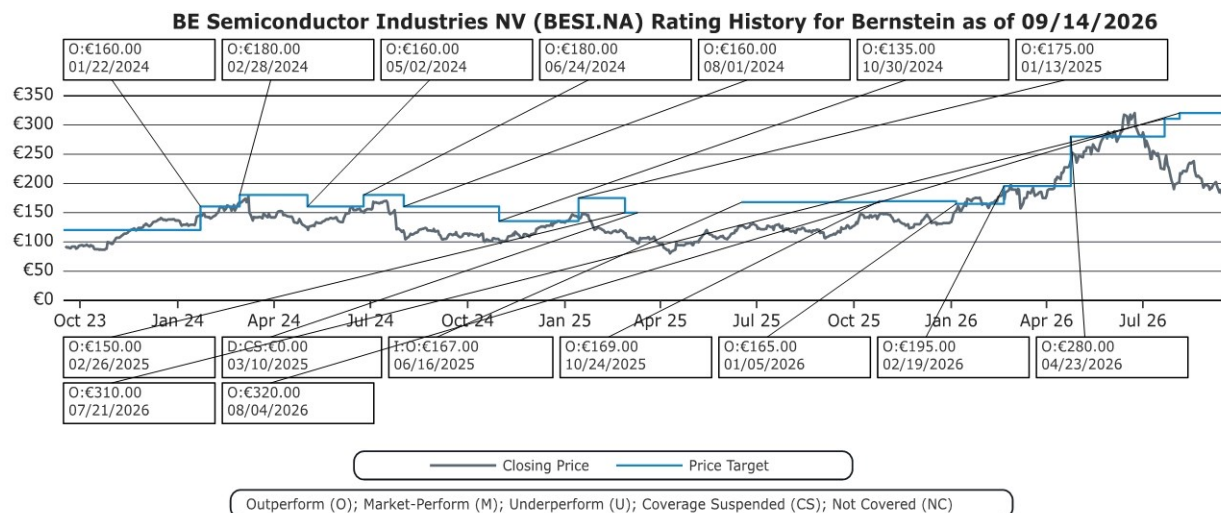
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Underperform	SELL	13.1%	13.6%

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